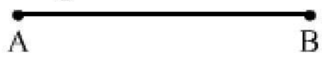
EXERCISE 6

Solution 1:

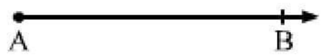
Axioms: An axiom is a basic fact that is taken for granted without proof.
i) The whole is greater than each of its parts.
Theorem: A statement that requires proof is called theorem.
ii) Pythagoras Theorem

Solution 2:

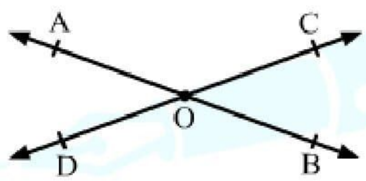
(i) Line segment: A line segment is a part of line that is bounded by two distinct end-points. A line segment has a fixed length.



(ii) Ray: A line with a start point but no end point and without a definite length is a ray.

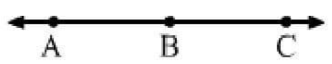


(iii) Intersecting lines: Two lines with a common point are called intersecting lines.

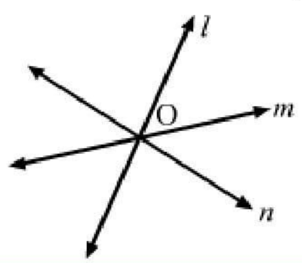


(iv) Parallel lines: Two lines in a plane without a common point are parallel lines.

(v) Half line: A straight line extending from a point indefinitely in one direction only is a half line.

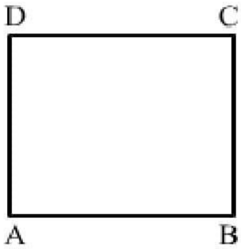


(vi) Concurrent lines: Three or more lines intersecting at the same point are said to be concurrent.

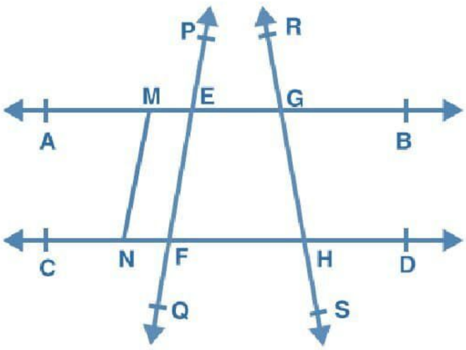


(vii) Collinear points: Three or more than three points are said to be collinear if there is a line, which contains all the points.

(viii) Plane: A plane is a surface such that every point of the line joining any two point on it, lies on it.



Solution 3:



(i) Points are A, B, C, D, P and R.

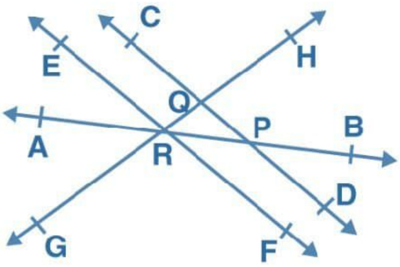
(ii) $\overline{EG}$, $\overline{FH}$, $\overline{EF}$, $\overline{GH}$, $\overline{MN}$

(iii) $\overrightarrow{EP}$, $\overrightarrow{GR}$, $\overrightarrow{HS}$, $\overrightarrow{FQ}$

(iv) $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$, $\overleftrightarrow{PQ}$, $\overleftrightarrow{RS}$

(v) Collinear points are M, E, G and B.

Solution 4:



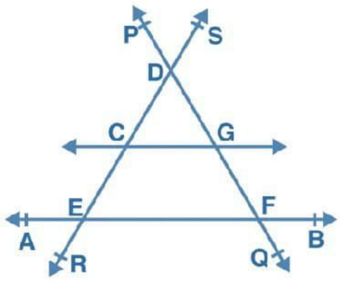
(i) $\overleftrightarrow{EF}$, $\overleftrightarrow{GH}$ points of intersection R and $\overleftrightarrow{AB}$, $\overleftrightarrow{CD}$, points of intersection P.

(ii) AB, EF, GH, point R

(iii) RB, RH, RF

(iv) $\overline{RQ}$ and $\overline{RP}$

Solution 5:



(a) Three lines are line AB, line PQ and line RS

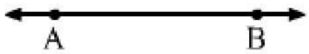
(b) CEFG

(c) A, E, F, B are four concurrent points.

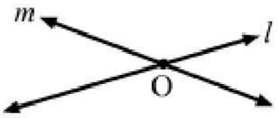
Solution 6:

(i) Infinite lines can be drawn through a given point.

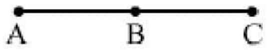
(ii) Only one line can be drawn through two given points.



(iii) Two lines at the most can intersect at one point only.



(iv) $\overline{AB}, \overline{BC}, \overline{AC}$



Solution 7:

(i) False. A line segment has a definite length.

(ii) False. A ray has one end-point.

(iii) False. A line has no definite length.

(iv) True

(v) False. $\overleftrightarrow{BA}$ and $\overleftrightarrow{AB}$ have different end-points.

(vi) True

(vii) True

(viii) True

(ix) True

(x) True

(xi) False. Two lines intersect at only one point.

(xii) True

Solution 8:

(i) It is given that L is the mid-point of AB.

$\therefore\ AL = BL = \frac{1}{2} AB \quad .....(1)$

Also, M is the mid-point of BC.

$\therefore\ BM = MC = \frac{1}{2} BC \quad .....(2)$

$AB = BC \quad$ (Given)

$\frac{1}{2}AB = \frac{1}{2}BC$

$\Rightarrow AL = MC \quad$ [From (1) and (2)]

(ii) It is given that L is the mid-point of AB.

$\therefore\ AL = BL = \frac{1}{2}AB$

$\Rightarrow 2AL = 2BL = AB \quad .....(3)$

Also, M is the mid-point of BC.

$\therefore BM = MC = \frac{1}{2} BC$

$\Rightarrow 2BM = 2MC = BC \quad .....(4)$

$BL = BM \quad$ (Given)

$\Rightarrow 2BL = 2BM$

$\Rightarrow AB = BC \quad$ [From (3) and (4)]

Multiple Choice Questions (MCQ)

Solution 1: (b) squares and circles

Solution 2: (b) public rituals

The construction of altars (or vedis) and fireplaces for performing vedic rituals resulted in the origin of the geometry of vedic period. Square and circular altars were used for household rituals whereas the altars with combination of shapes like rectangles, triangles and trapezium were used for public rituals.
Hence, the correct answer is option (b).

Solution 3: (c) nine

Solution 4: (b) 4 : 2 : 1

Solution 5: (a) 13

The famous treatise, '*The Elements*' by Euclid is divided into 13 chapters.
Hence, the correct answer is option (a).

Solution 6: (b) Greece

Solution 7: (c) Greece

Solution 8: (ii) Thales

Solution 9: (d) theorem

Solution 10: (a) a definition

Solution 11: (c) a postulate

Solution 12: (d) any polygon

Solution 13: (a) triangles

Solution 14: (c) 3

A solid shape has length, breadth and height. Thus, a solid has three dimensions.
Hence, the correct answer is option (c).

Solution 15: (b) 2

A plane surface has length and breadth, but it has no height. Thus, a plane surface has two dimensions.
Hence, the correct answer is option (b).

Solution 16: (a) 0

A point is a fine dot which represents an exact position. It has no length, no breadth and no height. Thus, a point has no dimension or a point has zero dimension.
Hence, the correct answer is option (a).

Solution 17: (c) surfaces

Solution 18: (b) curves

Solution 19: (d) 1

Solution 20: (d) universal truths in all branches of mathematics

Solution 21: (c) The floor and the ceiling of a room are parallel planes.

Solution 22: (c) If two circles are equal, then their radii are equal.

Solution 23: (c) Ray $\overrightarrow{AB}$ = Ray $\overrightarrow{BA}$

Solution 24: (c) C is an interior point of AB, such that AC = CB

Solution 25: (c) points A, C and B are collinear

Solution 26: (b) second

Euclid's second axiom states that if equals be added to equals, the wholes are equal.
$x + y = 15$
Adding z to both sides, we get
$x + y + z = 15 + z$

Thus, Euclid's second axiom illustrates the statement that when $x + y = 15$, then $x + y + z = 15 + z$.
Hence, the correct answer is option (b).

Solution 27: (a) First axiom

Euclid's first axiom states that the things which are equal to the same thing are equal to one another.
It is given that, the age of A is equal to the age of B and the age of C is equal to the age of B.
Using Euclid's first axiom, we conclude that the age of A is equal to the age of C.

Thus, Euclid's first axiom illustrates the relative ages of A and C.

Hence, the correct answer is option (a).